

Digital Telescope Module Specification

The Digital Telescope Module provides localized and global scanning capabilities for data embedded within the triangulation pipeline. It enables magnification, zooming, and deformation analysis across 2D→3D reconstructed data. This module is input-agnostic, designed to work with diverse datasets such as images, graphs, codebases, or astronomical data. Its core functionality is realized through three distinct lenses: Conformal, Eigenvector-based, and Beltrami.

1. Module Structure and Flow

The Digital Telescope module is composed of three lenses that can be applied independently or in sequence depending on the analysis goal. A Lens Manager orchestrates their application, taking as input a region-of-interest (ROI) defined on the dataset. Each lens produces a transformation map that is applied to the ROI, yielding a zoomed or reshaped representation.

2. Lenses

2.1 Conformal Lens

The Conformal Lens preserves local angles during magnification. It is based on conformal mapping methods, such as Möbius transformations or Riemann mapping techniques. This ensures that shapes are magnified without introducing angular distortion, making it ideal for geometry-preserving local zooming. Structure & Functions: - Input: ROI (points, polygons, textures). - Transformation: Apply conformal map $T(z) = (az+b)/(cz+d)$. - Output: Magnified ROI with angular fidelity.

2.2 Eigenvector Lens

The Eigenvector Lens leverages eigen-decomposition of adjacency or Laplacian matrices derived from the ROI. It provides magnification along dominant eigen-directions, highlighting structural or spectral properties within the data. This is useful for highlighting latent patterns, clusters, or oscillations. Structure & Functions: - Input: ROI adjacency/Laplacian matrix. - Transformation: Compute eigenpairs (λ, v) , apply scaling along chosen eigenvectors. - Output: Deformed ROI reflecting spectral magnification.

2.3 Beltrami Lens

The Beltrami Lens uses a Linear Beltrami Solver (LBS) to allow controlled, global+local deformations while preserving quasiconformal properties. It introduces a complex dilation parameter μ that governs distortion, enabling zoom that balances between global context and local magnification. Structure & Functions: - Input: ROI mesh/grid, Beltrami coefficient $\mu(x,y)$. - Transformation: Solve $\partial f/\partial \bar{z} = \mu \partial f/\partial z$ via LBS. - Output: Quasiconformally magnified ROI, smoothly integrated with the global dataset.

3. Lens Manager

The Lens Manager orchestrates the application of the three lenses. It allows: - Selecting lens type(s). - Adjusting parameters (zoom factor, eigenvalue index, Beltrami coefficient). - Composing multiple lenses sequentially (e.g., conformal zoom followed by eigenvector deformation). - Exporting transformation maps for further processing in triangulation or visualization.

4. Outputs and Integration

Outputs of the Digital Telescope include transformed meshes, magnified regions, and transformation maps. These outputs can be: - Passed back into the triangulation pipeline for refinement. - Used for visualization and exploration. - Exported as standalone magnified models for analysis.